

**Cover Letter:**

August 24, 2026

*X PhD Residency- AI for Chemistry*

Dear X AI Residency Team,

I am writing to express my interest in the AI Residency position. I am a M.Sc. in Physics and Ph.D. candidate in Chemistry at Northwestern University, co-advised by Prof. Monica Olvera de la Cruz and Prof. Maria K. Chan at Argonne National Laboratory, where I develop computational tools to understand charge transport in complex materials at the intersection of machine learning, quantum chemistry, and molecular dynamics.

One problem that motivates my research is the growing energy cost of AI systems. Neuromorphic hardware, devices that process information through the coupled movement of ions and electrons, the way biological neurons do, offers a fundamentally different and potentially far more energy-efficient approach to computing. Understanding and designing these materials from the atomistic level up is what my PhD is ultimately about, and it is the kind of challenge that keeps me motivated.

My current work focuses on developing machine learning interatomic potentials and simulation frameworks for mixed ionic-electronic materials relevant to neuromorphic computing, integrating quantum mechanical calculations, and molecular simulations, to capture how atomistic interactions influence macroscopic electronic behavior. Much of this has happened in close collaboration with experimental groups through the Northwestern Materials Research Science and Engineering Center, where I have used simulations to interpret measurements and guide materials discovery such as developing ML screening approaches to identify ionic liquids with high charge transport and sufficient infrared activity for FTIR characterization, studying the exfoliation of inorganic materials relevant to printed field-effect transistors, and calculating electronic excitations in organic molecules to compare directly with ultrafast spectroscopy experiments. This work has taught me that the most useful simulation is one that speaks the same language as the measurement.

Before my Ph.D., I worked at Mexico's national research council (CONAHCYT, now Secihti), where I built computational tools to automate the evaluation of federally funded research projects, an experience that showed me how the same data-driven thinking I apply to molecules can be useful in very different contexts. It also sparked a growing interest in how large language models and agentic AI systems might work alongside scientific workflows, something I find increasingly relevant as these tools become more capable.

At this stage of my career, I am eager to bring the experience I have built during my Masters and PhD into a new environment and see how it translates to a different kind of scientific challenge, one driven by product thinking and real-world deployment. My background has given me a broad range of experience at the intersection of computational simulations, and chemistry, and I want to continue developing it alongside researchers who think with a moonshot perspective. I believe that experience would shape not just my technical skills but the scientist I become.

Thank you for your time and consideration.

Warmly,

**Angel Ivan Rodriguez-Leon**